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SE-AIML

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Experiment 8: Clustering of Countries with World Bank API

Part 1: Data Preprocessing

Import libraries

In [43]:
```
import requests
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans,AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram,linkage
```

Retrieve Data from api

In [44]:
```
def fetch_indicator(code, year="2022"):
    url = f"https://api.worldbank.org/v2/country/all/indicator/{code}?date={year}&format=jsona
    response = requests.get(url).json()

    data = {}

    for item in response[1]:
        country = item["country"]["value"]
        value = item["value"]
        if value is not None:
            data[country] = value

    return pd.Series(data, name=code)
```

In [45]:
```
gdp=fetch_indicator("NY.GDP.PCAP.CD")
pop=fetch_indicator("SP.POP.TOTL")
life=fetch_indicator("SP.DYN.LE00.IN")
lit=fetch_indicator("SE.ADT.LITR.ZS")
```

Create Dataset and Clean the data

In [46]:
```
df=pd.concat([gdp,pop,life,lit],axis=1)
df.columns=["gdp","pop","life","lit"]
df.dropna(inplace=True)
df.head()
```

Out[46]:

	gdp	pop	life	lit
Africa Eastern and Southern	1679.327622	731821393	64.487020	73.055977
Africa Western and Central	2138.473153	497387180	57.987813	60.780979
Arab World	7950.355820	471352066	71.876096	78.271927
Central Europe and the Baltics	19530.155323	100071871	76.652299	99.479919
Early-demographic dividend	4043.827763	3470644834	71.430455	81.242798

In [47]:
```
df.tail()
```

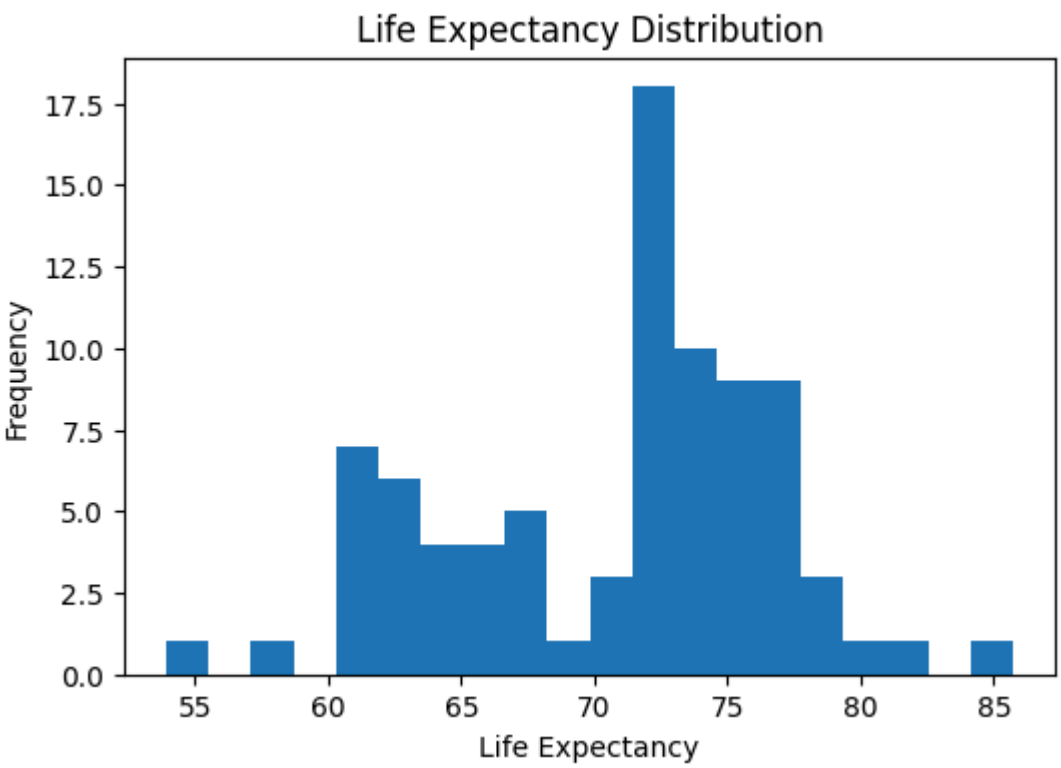
Out[47]:

	gdp	pop	life	lit
Uruguay	20818.614609	3390913	76.468	98.849998
Uzbekistan	2578.666894	34938955	72.147	100.000000
Viet Nam	4147.697772	99680655	74.502	96.129997
West Bank and Gaza	3799.955270	5043612	76.662	97.839996
Zambia	1447.123101	20152938	65.279	79.975143

Data Scaling

In [48]:
```
scalar=StandardScaler()
X_scaled=scalar.fit_transform(df)
```

In [49]:
```
plt.figure(figsize=(6,4))
plt.hist(df["life"],bins=20)
plt.xlabel("Life Expectancy")
plt.ylabel("Frequency")
plt.title("Life Expectancy Distribution")
plt.show()
```



Part 2: Elbow method

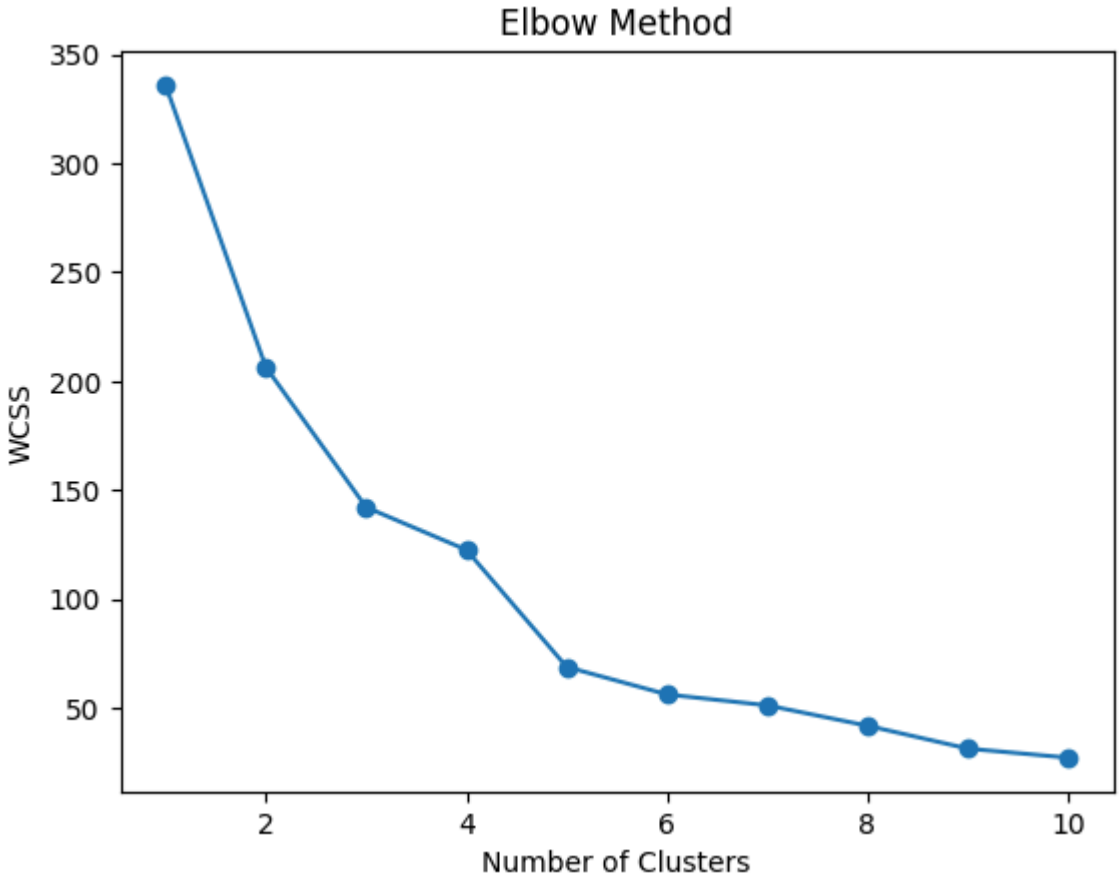
In [50]:
```
wcss=[]

for k in range(1,11):
    km = KMeans(n_clusters=k,random_state=0)
    km.fit(X_scaled)
    wcss.append(km.inertia_)

print("k\tWCSS")
for i in range(10):
  print(i+1,"\t",wcss[i])

plt.plot(range(1,11),wcss,marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

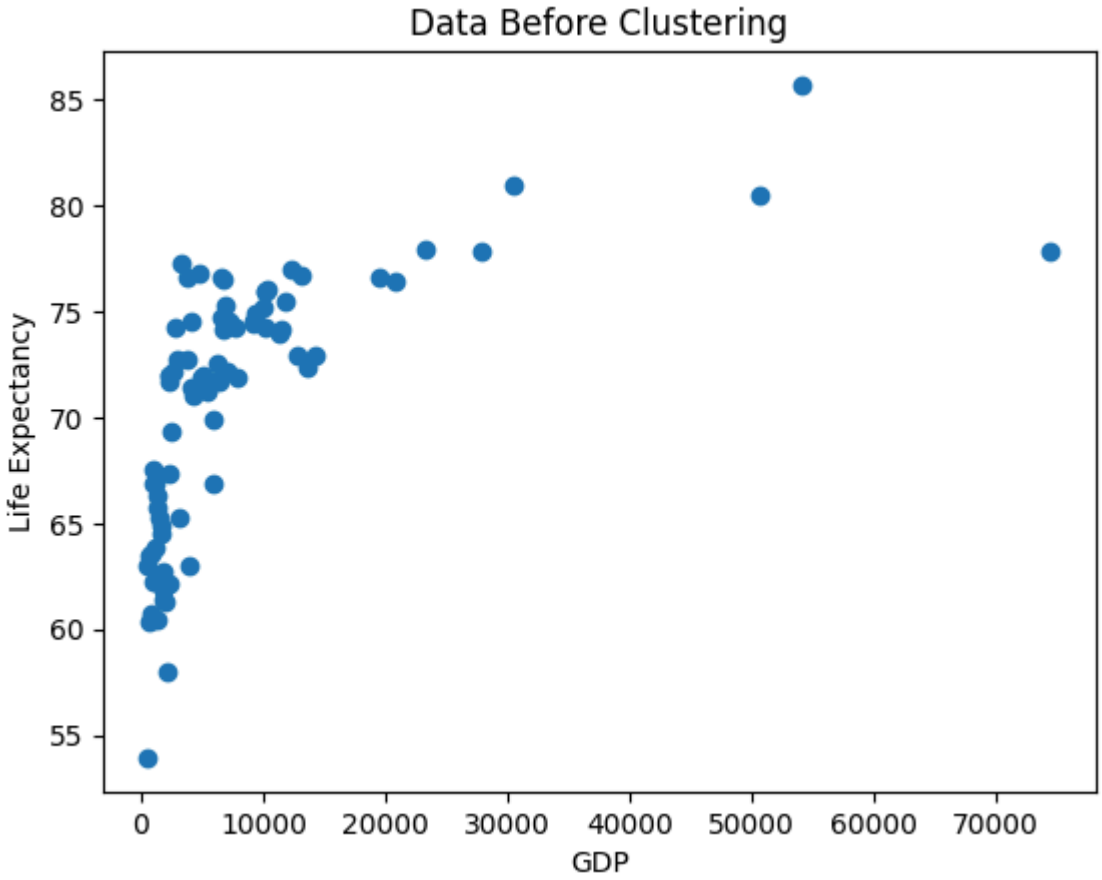
```
k       WCSS
1        336.00000000000006
2        206.43717192603813
3        142.060283249676
4        122.4640064142117
5        68.7123470772603
6        56.09271840581914
7        51.08265517326419
8        41.70851027024396
9        31.213863029560212
10       27.04725098650772
```



Part 3: K-Means Clustering

In [51]:
```
plt.scatter(df['gdp'],df['life'])
plt.xlabel("GDP")
```

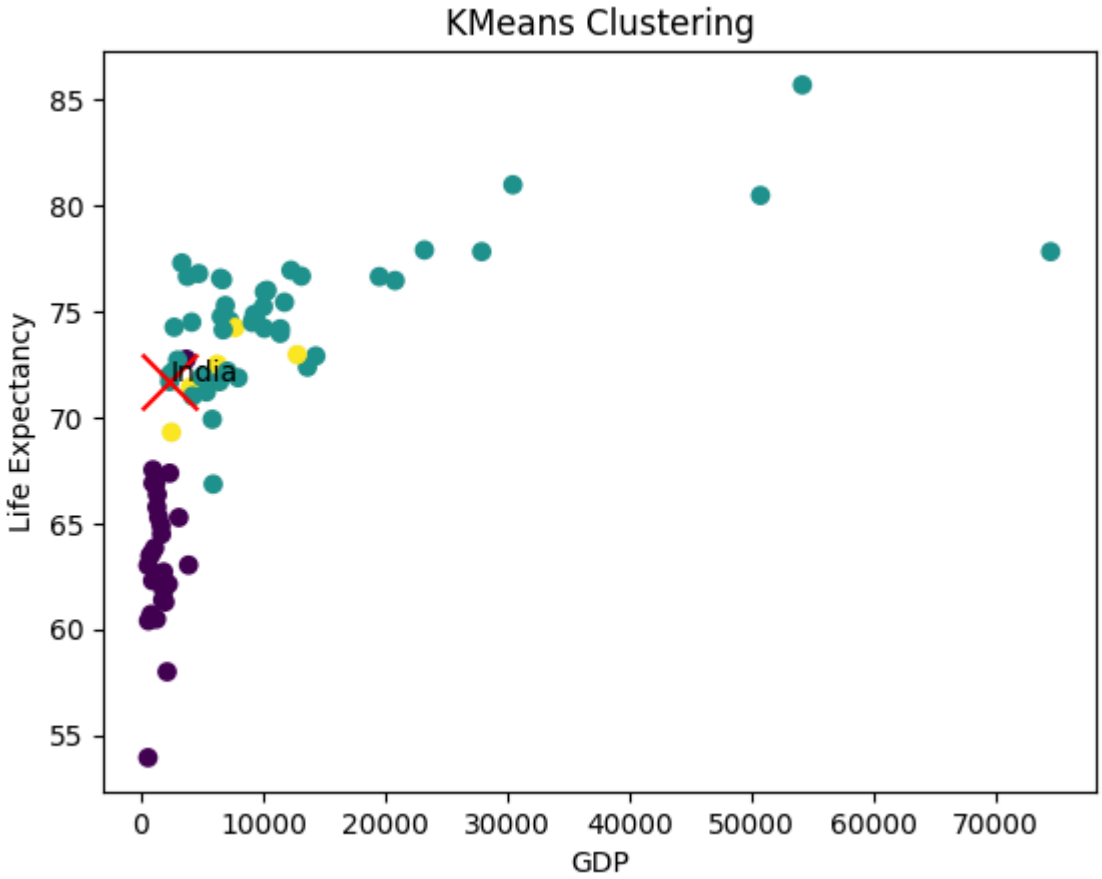
```
plt.ylabel("Life Expectancy")
plt.title("Data Before Clustering")
plt.show()
```



In [52]:
```
kmeans=KMeans(n_clusters=3,random_state=0)
labels_km=kmeans.fit_predict(X_scaled)
df["KMeanscluster"]=labels_km
```

In [53]:
```
plt.scatter(df['gdp'],df['life'],c=labels_km)
plt.xlabel("GDP")
plt.ylabel("Life Expectancy")
plt.title("KMeans Clustering")

india=df.loc["India"]
plt.scatter(india["gdp"],india["life"],s=400,marker="x",color="red")
plt.text(india["gdp"],india["life"],"India")
plt.show()
```

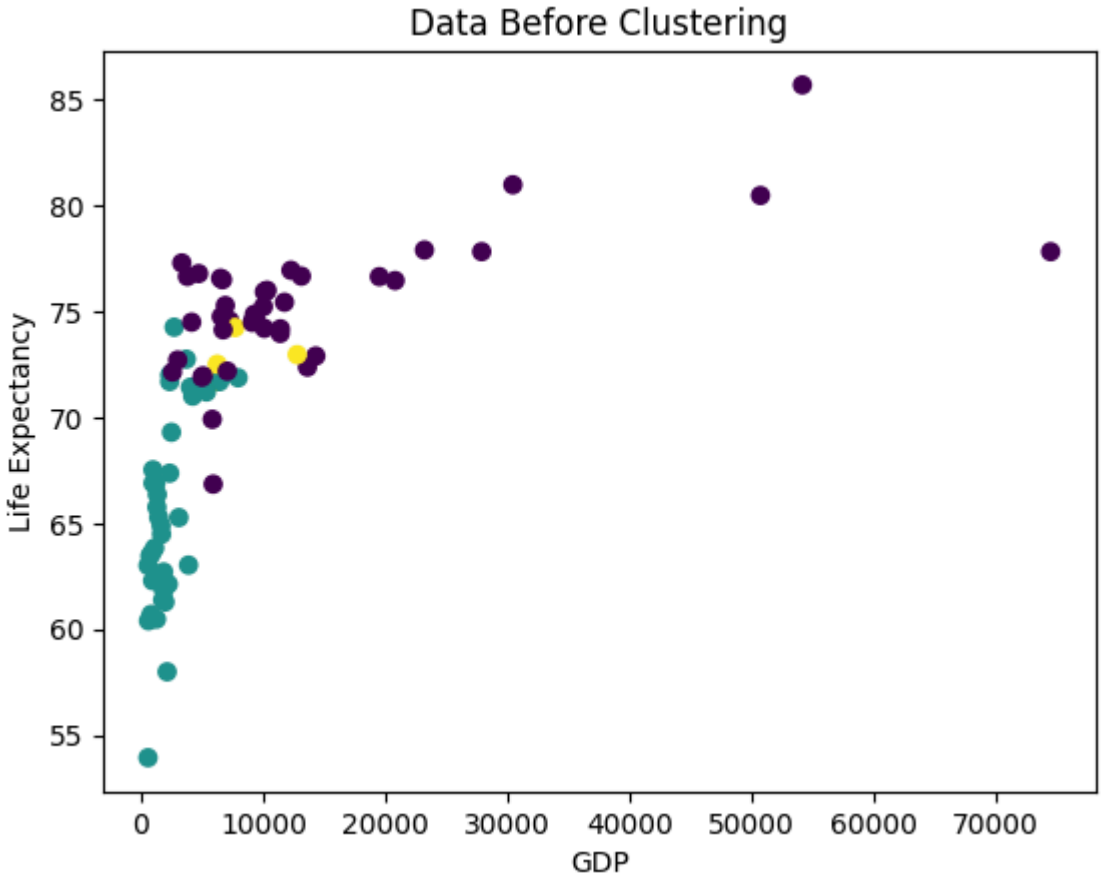


Part 4: Agglomerative Clustering

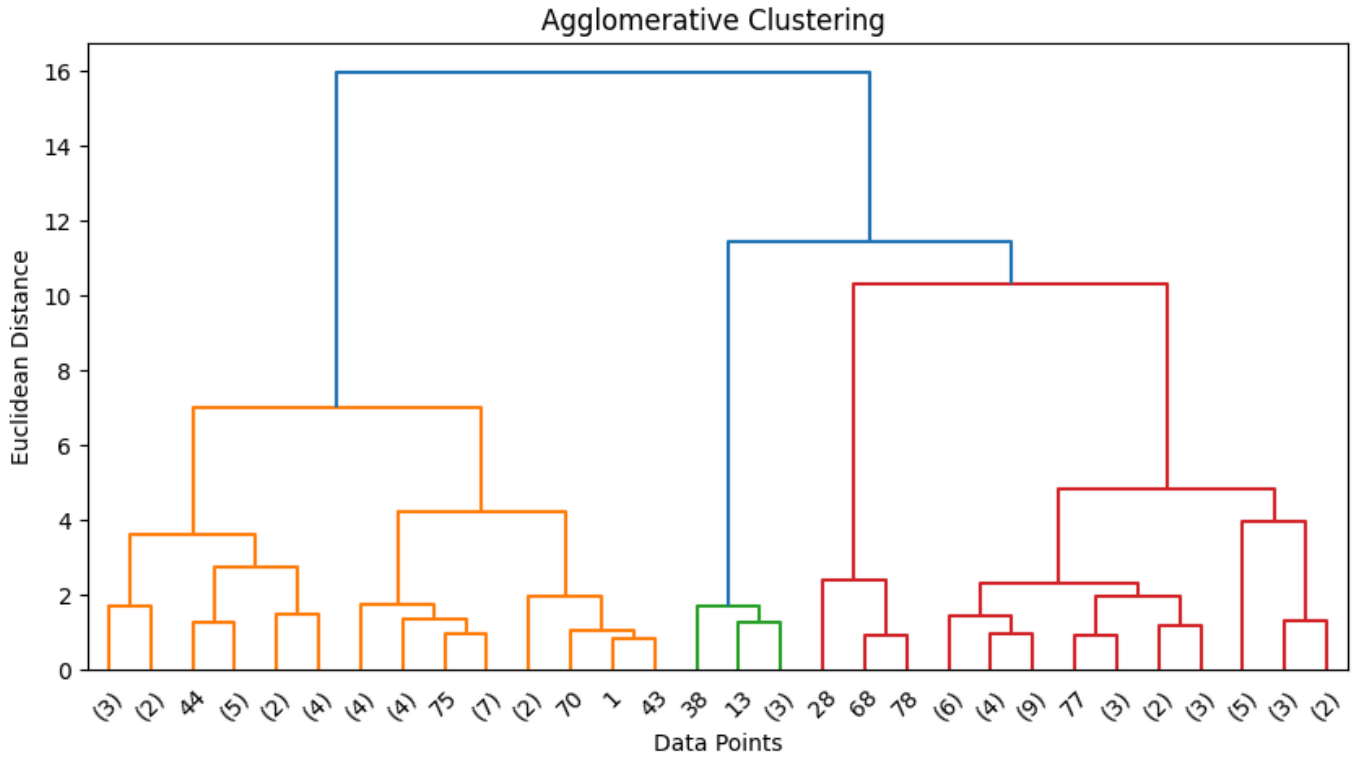
In [54]:
```
hc=AgglomerativeClustering(n_clusters=3)
labels_hc=hc.fit_predict(X_scaled)

df["HierarchialCluster"]=labels_hc
```

In [55]:
```
plt.scatter(df['gdp'],df['life'],c=labels_hc)
plt.xlabel("GDP")
plt.ylabel("Life Expectancy")
plt.title("Data Before Clustering")
plt.show()
```



In [56]:
```
Linked=linkage(X_scaled,method="ward")
plt.figure(figsize=(10,5))
dendrogram(Linked,truncate_mode='lastp',p=30)
plt.title("Agglomerative Clustering ")
plt.xlabel("Data Points")
plt.ylabel("Euclidean Distance")
plt.show()
```



Part 5: Result

Table 1

In [57]:
```
country_name="India"

result=df.loc[country_name]
print("Country:",country_name)
print("GDP:",result["gdp"])
print("Population:",result["pop"])
print("Life Expectancy:",result["life"])
print("Literacy Rate:",result["lit"])
print("KMeans Cluster:",result["KMeanscluster"])
print("Hierarchial Cluster:",result["HierarchialCluster"])
```

```
Country: India
GDP: 2347.44829434623
Population: 1425423212.0
Life Expectancy: 71.698
Literacy Rate: 76.3199996948242
KMeans Cluster: 1.0
Hierarchial Cluster: 1.0
```

In [58]:
```
country_name="North America"

result=df.loc[country_name]
print("Country:",country_name)
print("GDP:",result["gdp"])
print("Population:",result["pop"])
print("Life Expectancy:",result["life"])
print("Literacy Rate:",result["lit"])
print("KMeans Cluster:",result["KMeanscluster"])
print("Hierarchial Cluster:",result["HierarchialCluster"])
```

```
Country: North America
GDP: 74535.6365392629
Population: 373018004.0
Life Expectancy: 77.8327679594239
Literacy Rate: 98.7891464233398
KMeans Cluster: 1.0
Hierarchial Cluster: 0.0
```

In [59]:
```
country_name="Brazil"

result=df.loc[country_name]
print("Country:",country_name)
print("GDP:",result["gdp"])
print("Population:",result["pop"])
print("Life Expectancy:",result["life"])
print("Literacy Rate:",result["lit"])
print("KMeans Cluster:",result["KMeanscluster"])
print("Hierarchial Cluster:",result["HierarchialCluster"])
```

```
Country: Brazil
GDP: 9281.33282136864
Population: 210306415.0
Life Expectancy: 74.872
Literacy Rate: 94.3853874531348
KMeans Cluster: 1.0
Hierarchial Cluster: 0.0
```

In [60]:
```
country_name="Viet Nam"

result=df.loc[country_name]
print("Country:",country_name)
print("GDP:",result["gdp"])
print("Population:",result["pop"])
print("Life Expectancy:",result["life"])
print("Literacy Rate:",result["lit"])
```

```
print("KMeans Cluster:",result["KMeanscluster"])
print("Hierarchial Cluster:",result["HierarchialCluster"])
```

```
Country: Viet Nam
GDP: 4147.69777213621
Population: 99680655.0
Life Expectancy: 74.502
Literacy Rate: 96.129997253418
KMeans Cluster: 1.0
Hierarchial Cluster: 0.0
```

In [61]:

```
country_name="Bahrain"

result=df.loc[country_name]
print("Country:",country_name)
print("GDP:",result["gdp"])
print("Population:",result["pop"])
print("Life Expectancy:",result["life"])
print("Literacy Rate:",result["lit"])
print("KMeans Cluster:",result["KMeanscluster"])
print("Hierarchial Cluster:",result["HierarchialCluster"])
```

```
Country: Bahrain
GDP: 30470.5219276023
Population: 1524693.0
Life Expectancy: 80.992
Literacy Rate: 97.870002746582
KMeans Cluster: 1.0
Hierarchial Cluster: 0.0
```

Table 3

In [62]:

```
cluster_summary = df.groupby("KMeanscluster")[["gdp","pop","life","lit"]].mean()

print(cluster_summary)
```

```
                        gdp           pop       life        lit
KMeanscluster
0               1601.971896  4.854521e+08  63.478867  66.779721
1              12288.225626  4.669122e+08  74.753258  92.910908
2               6437.349751  5.501554e+09  71.975830  84.851687
```